TOUR CEYLON: AI-Based Tourism Risk Prediction & Advisory Platform for Sri Lanka Tourism

INDIVIDUAL THESIS

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ABSTRACT

Tourist sites in Sri Lanka can become overcrowded, yet visitor figures for individual sites are published only as annual totals, so no site-level daily crowd record exists for building or testing a prediction service. This thesis presents the crowd and risk prediction component of Tour Ceylon, a four-component tourism safety platform, and asks how well a daily crowd index can be predicted for sites a model has never seen. Following a design science approach, I assembled a dataset of 197 sites and 144,007 daily records (2024–2025), defined a constructed crowd index as the prediction target, and found and disclosed eleven data defects. A Random Forest regressor was evaluated on a primary hold-out of ten entire sites, site-grouped cross-validation over all 197 sites, a 2024-to-2025 temporal split, and random and unshuffled splits kept only as optimistic references. On the ten unseen sites the model achieved MAE 0.1232 and R^2 0.3331, against 0.2835 and −0.3476 for a mean predictor and 0.2179 and 0.1201 for a category-and-month mean. Site-grouped cross-validation gave MAE 0.1159 and R^2 0.5071, whereas the random-split R^2 of 0.8492 largely reflects sites seen in training. Ablation showed that capacity per day carries most of the predictive signal, while hotel-occupancy and flight features added none. Error concentrated in a few sites: Wilpattu National Park alone contributed about 46% of the summed per-site error of the ten held-out sites. Because the target is constructed rather than observed, the results show how well the index is reproduced, not real crowding. I built the dataset, the model, the validation protocol and the API for this component. The work contributes a documented dataset, a leakage-aware evaluation protocol and an explicit account of its validity limits

Keywords: crowd prediction, Random Forest, site-holdout validation, tourism risk, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CV	Cross-Validation
DSR	Design Science Research
MAE	Mean Absolute Error
R^2	Coefficient of Determination
REST	Representational State Transfer
RF	Random Forest
SD	Standard Deviation
SDG	Sustainable Development Goal
SLTDA	Sri Lanka Tourism Development Authority
SOS	Emergency distress alert (the Tour Ceylon emergency component)
SUS	System Usability Scale
UAT	User Acceptance Testing
UNESCO	United Nations Educational, Scientific and Cultural Organization

1 INTRODUCTION

1.1 Overall project context

Tour Ceylon (Group R26-IT-152) is a tourism platform for Sri Lanka built by four students. Each student owns one component: an SOS emergency system (De Silva D.S.K), an itinerary generator (Wanniarachchi P.M.R), a wellness and Ayurveda recommender (De Silva D.C.M), and the crowd and risk prediction service reported here. The shared problem, architecture and integration are described in the Common Integrated Solution Report. This thesis describes only the fourth component and its interfaces.

1.2 Individual component context

The proposal described this component as a "Tourism Data Analyzer" that would collect site and hotel-occupancy data, track flight volumes, find overcrowding patterns, predict safe busy times and update itineraries. During development it became a crowd and risk prediction service, the Tourism Risk & Context Intelligence System. For a chosen site and date, the service returns a crowd score between 0 and 1 and a risk advisory that other components can use.

1.3 Problem statement

A traveller or itinerary planner who wants to avoid overcrowded sites needs an estimate of how busy a site will be on a given day. In Sri Lanka no site-level daily record exists: site-level visitor statistics are published as annual aggregates by the Sri Lanka Tourism Development Authority (SLTDA) [1]. A prediction service therefore cannot be trained on observed daily crowds. It must be trained on a constructed index, and it must still be shown to work on sites it has never seen. The problem is to build and honestly evaluate such a service.

1.4 Research gap

A common way to evaluate a model on data with repeated measurements per site is a random split, in which days from the same site appear in both training and test sets.

Roberts et al. [2] show that this gives over-optimistic error when observations are grouped, and they recommend blocking by group. The gap addressed here is an evaluation of daily crowd prediction on entirely unseen sites, using a documented and defect-audited dataset, with the limits of a constructed target stated openly.

1.5 Aim and objectives

Aim. To design, build and evaluate a crowd and risk prediction service for Sri Lankan tourist sites, and to report honestly how well it generalises to unseen sites.

Objectives.

1. Assemble a 197-site daily dataset and document its defects.
2. Define a constructed daily crowd index as the prediction target.
3. Train a Random Forest and compare it with simple baselines.
4. Evaluate on unseen sites and unseen time periods, with leakage checks.
5. Expose predictions to the platform through a Representational State Transfer (REST) application programming interface (API) under /api/risk.
6. Assess the usefulness of the service with users.

1.6 Research questions or hypotheses

- **RQ1.** How accurately can a Random Forest predict a constructed daily crowd index for sites it has never seen, compared with simple baselines?
- **RQ2.** Which features carry the predictive signal, and which contribute nothing measurable?
- **RQ3.** How far do random-split results overstate performance relative to site-holdout and site-grouped evaluation?

1.7 Scope limitations and assumptions

The dataset covers 197 sites and 144,007 daily records for 2024–2025, in 28 columns. The following limitations apply and are examined in Chapters 3 and 7:

- The crowd index is constructed, not observed.
- Capacity per day is derived from annual visitor figures for 111 sites.
- Weather depends only on the month.
- Hotel occupancy and flight arrivals are identical across sites on any given date.
- The 2024 festival dates could not be verified.
- The serving application derives the public-holiday and festival flags from one shared check, whereas training used separate flags. Removing the weekend and public-holiday flags changed site-holdout mean absolute error (MAE) by 0.0004, and the festival flag was not tested separately.
- No field validation of the index against observed crowds has been carried out.

Table 1.1: Proposed scope and what was built

Proposed task or novelty	What was built
Collect sites and hotel-occupancy data	A 197-site dataset with date-level occupancy, identical across sites
Track flight tourist volumes	A date-level flight series, identical across sites; dropping it changed site-holdout MAE from 0.1232 to 0.1212 (Table 6.4)
Find overcrowding patterns	A constructed daily crowd index and a Random Forest that predicts it
Predict safe busy times	A daily crowd score per site through the `/api/risk` endpoints
Update itineraries dynamically	The endpoint `/api/risk/itinerary-check`
Flight-based crowd forecast	Not implemented as proposed, because flight features carry no measurable signal
Green-site focus	The endpoint `/api/risk/green-sites`, which ranks 114 eco-sites
Improves from real trips	Not implemented; listed in Section 8.4

1.8 Individual contribution

I created and audited the dataset, defined the crowd index, trained and evaluated the model, wrote the validation script validate_model.py, and exposed the model through the /api/risk endpoints. The other three components were built by my teammates and are described in their theses.

1.9 Thesis structure

Chapter 2 reviews related work. Chapter 3 sets out the methodology and the dataset. Chapter 4 covers the design. Chapter 5 covers implementation, integration and testing. Chapter 6 reports results and validation. Chapter 7 discusses them and their limits. Chapter 7 discusses them and their limits. Chapter 8 concludes and gives future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review is targeted, not systematic. It covers the sources needed to justify four decisions: the research approach, the model family, the evaluation design and the platform context. It does not claim to survey all work on crowd forecasting.

2.2 Domain and theoretical foundations

Design science. Hevner et al. [3] describe design science as the creation and evaluation of artefacts that solve identified problems. Peffers et al. [4] give a design science research (DSR) process of problem identification, objectives, design and development, demonstration, evaluation and communication. This thesis follows that sequence: problem and objectives in Chapter 1, design in Chapter 4, development in Chapter 5, and evaluation in Chapters 6 and 7.

Random Forest. A Random Forest (RF) [5] averages many decision trees, each trained on a bootstrap sample with a random subset of features at each split. For regression it returns the mean of the trees. It handles non-linear interactions and mixed feature types without feature scaling. Averaging reduces variance but does not remove bias or leakage from the way data are split, a point developed by Hastie et al. [6]. The model was implemented in scikit-learn [7].

Validation under grouped data. Each site contributes 731 daily rows, so rows from one site are strongly related. Roberts et al. [2] show that random cross-validation (CV) then reports a model's skill at reproducing sites it has already seen. Blocking the folds by site gives an estimate for new sites. This is why site-holdout evaluation is the primary design here.

2.3 Related research

Buhalis and Amaranggana [8] describe smart tourism destinations, in which information and communication technologies personalise services for visitors. The Tour Ceylon platform belongs to this setting, and a crowd-aware advisory is one

such service. The evaluation argument of this thesis rests on [2] and not on any claim about other crowd-forecasting results.

2.4 Existing systems products and techniques

Site-level visitor statistics in Sri Lanka are published as annual aggregates [1]. They cannot say how busy a particular site is on a particular day. That limitation is the reason this project constructs an index and then evaluates how well it can be predicted.

2.5 Gap synthesis and conceptual framework

Table 2.1 links each issue to the response taken in this thesis.

Table 2.1: From gap to design response

Issue	**What the sources establish**	**Response in this thesis**
No observed daily site-level crowd data	Site-level statistics are annual only [1]	A constructed index, openly disclosed as such
Optimistic random-split evaluation	Grouped data need blocked validation [2]	Site-holdout as the primary evaluation, with grouped CV
Model choice	RF averages trees and handles non-linearity [5]	RF compared with Ridge and simple means

The conceptual framework is: constructed index → leakage-aware evaluation on unseen sites → honest statement of what the result does and does not show.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows DSR [3], [4]. The artefact is a crowd prediction service. Evaluation is quantitative and uses several validation designs, so that each claim names the split it came from.

3.2 Data participants or study materials

The dataset contains 197 sites and 144,007 daily records from 1 January 2024 to 31 December 2025 (72,102 rows in 2024 and 71,905 in 2025), in 28 columns. The target is a number between 0 and 1: mean 0.4359, standard deviation (SD) 0.2436, with 3.9% of rows at 1.0 and 0.4% at 0.0. The mean was 0.4355 in 2024 and 0.4364 in 2025. Rows are ordered by date, then site. The index is defined in Appendix A.1.

For the primary evaluation, 187 sites (136,697 rows) were used for training and 10 sites (7,310 rows) were held out entirely: Bentota Beach, Bundala National Park, Colombo Lotus Tower, Kalpitiya Beach, Kande Viharaya, Kandy Lake, Minneriya National Park, Pasikuda Beach, Tangalle Beach and Wilpattu National Park. The model uses 16 input features (43 after one-hot encoding), including site capacity per day, calendar and season variables, weekend and holiday flags, weather, hotel occupancy, flight arrivals and site category.

Eleven data defects were identified and disclosed (Table 3.1). No personal data were collected.

Table 3.1: Data defects

#	**Defect**	**Handling**
1	Entrance fee recorded as −1 at 17 sites	Corrected
2	11 duplicate sites (208 → 197)	Merged
3	Festival flag inactive because of a year mismatch	Corrected
4	Multi-day festivals matched on exact dates	Corrected for the 2025

#	Defect	Handling
		Perahera; 2024 dates unverifiable
5	UNESCO (United Nations Educational, Scientific and Cultural Organization) sites under-flagged (7 → 13)	Corrected
6	Moneragala / Monaragala spelling	Unified
7	Site IDs with gaps	Re-indexed 1–197
8	Capacity derived from annual visitors for 111 sites	Disclosed
9	Weather depends only on the month	Disclosed
10	Hotel occupancy identical across sites per date	Disclosed
11	Flight arrivals identical across sites per date	Disclosed

3.3 Experimental or study procedure

All results come from one script, validate_model.py, run in a Python 3.10.18 environment with scikit-learn 1.7.1, pandas 2.3.2 and NumPy 2.2.6. Before any new analysis, the script re-computes the locked primary result and stops if it differs. It reproduced MAE 0.1232 and R^2 0.3331 exactly. Its read-only check confirmed that the model and data folders were unchanged. The Random Forest used n_estimators = 50, max_depth = 12 and min_samples_split = 7.

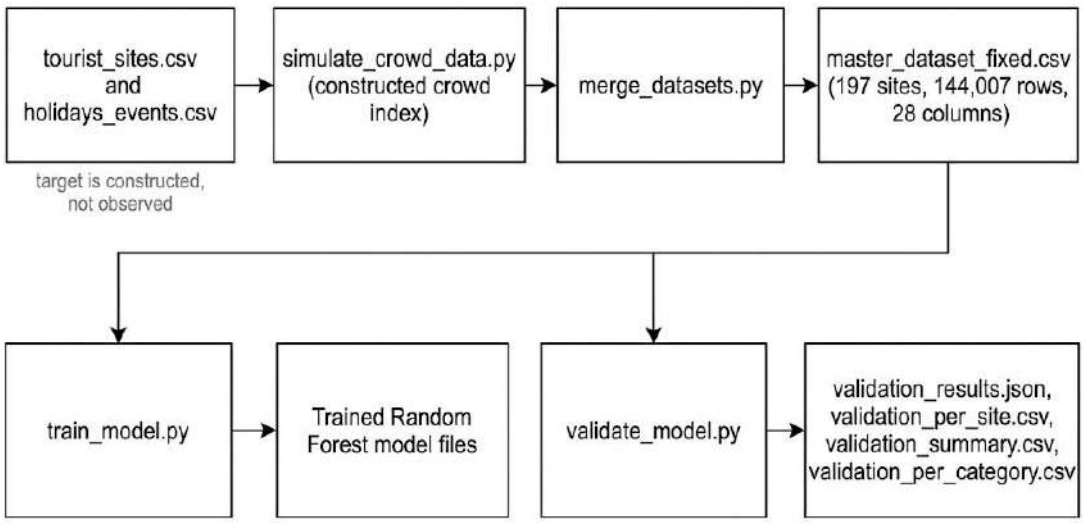


Figure 3.1 shows the pipeline from the raw inputs to the validated results.

The designs were:

1. **Site-holdout (primary):** ten sites unseen.
2. **Site-grouped CV:** five folds × three repeats over all 197 sites.
3. **Temporal:** train on 2024, test on 2025, for the same sites, for the ten held-out sites and for site-grouped folds.
4. **Random 80/20 and unshuffled 5-fold:** reference only, since sites are seen in training (Appendix A.2).
5. **Baselines:** mean, category mean, category-and-month mean, site mean, site-and-month mean and Ridge regression.
6. **Ablation and shuffled-target check.**
7. **Leakage-free tuning check:** 27 settings scored by site-grouped CV on the 187 training sites only (Appendix A.5).

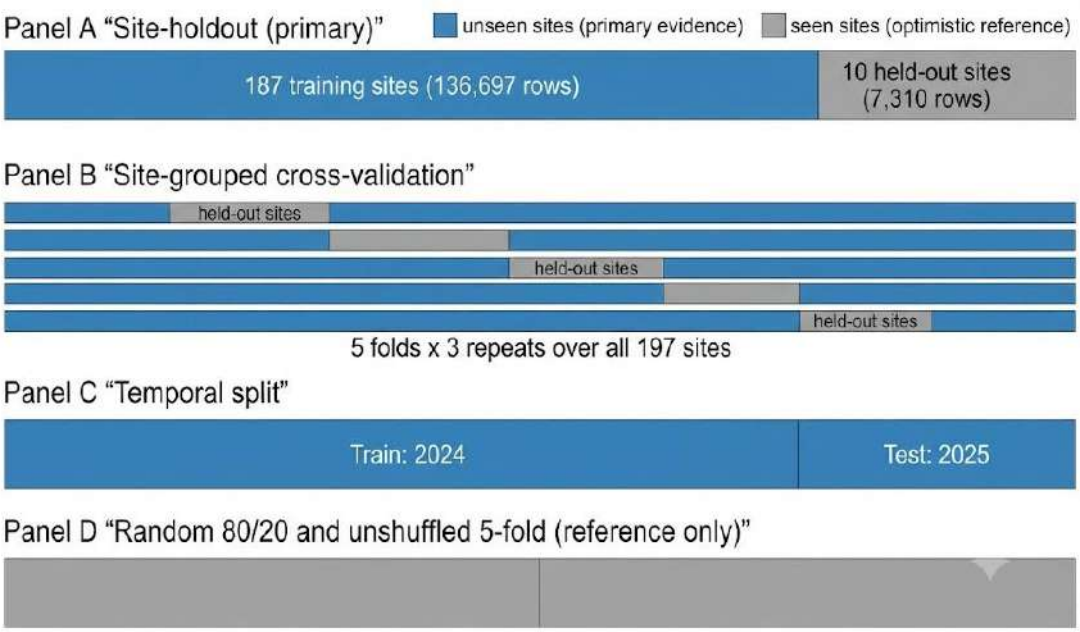


Figure 3.2: Validation designs (unseen-site designs are the primary evidence)

3.4 Measures and evaluation metrics

MAE and the coefficient of determination (R^2) are reported for regression. Accuracy and balanced accuracy are reported for the three-class Low/Medium/High classification. Every table names its split.

3.5 Data analysis

Results are reported as mean ± sample SD across folds. Models are compared with baselines on the same split. Errors are analysed per site and per category using out-of-fold predictions, and calibration is checked by deciles of predicted score.

3.6 Ethics privacy security and responsible AI

The dataset describes sites, not people. Advisory output is presented as an estimate of a constructed index, not as observed crowding. Generative artificial intelligence (AI) assistance is disclosed in Appendix B.

3.7 Reproducibility and research artefacts

The validation script writes validation_results.json, validation_summary.csv, validation_per_site.csv, validation_per_category.csv and a comparison figure. It backs up earlier outputs before overwriting. The saved outputs of the run (8.8 minutes), together with the model metadata file, the confusion matrix produced by the training script and the dataset, are the source of every number in Chapters 6 and 7.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The service must return a daily crowd score for a site and date, support an itinerary-level risk check, rank green sites, and return a clear "not covered" response for destinations outside the 197-site model. Non-functional requirements are evidence-based accuracy on unseen sites (Chapter 6), response time (Chapter 5) and clear error handling when a site is not covered.

4.2 Component architecture within the complete solution

The risk service is a Flask application whose routes are registered under /api/risk in the unified Tour Ceylon backend (port 5002) by risk_mount.py. A React frontend for the risk component runs on port 5182. The SOS tourist frontend calls the risk endpoints.

Figure 4.1 shows how the risk service sits inside the unified backend.

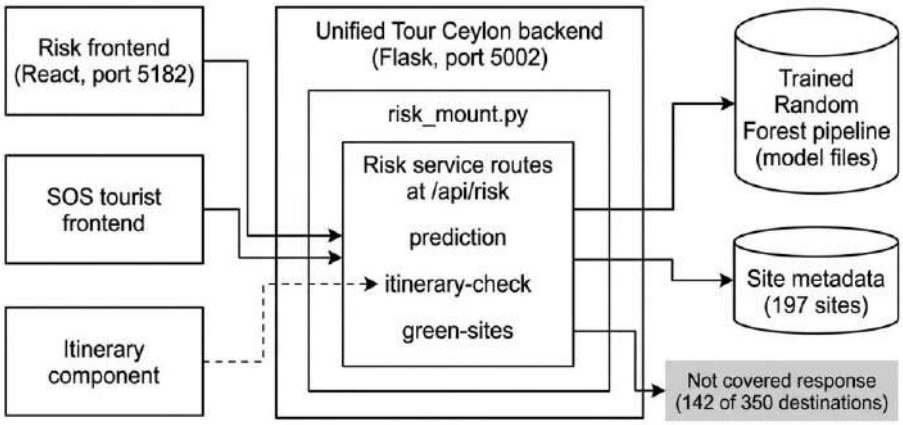


Figure 4.1: Architecture of the crowd and risk component inside Tour Ceylon

4.3 Design alternatives and decisions

Table 4.1 lists the main design decisions. The tuning evidence for the hyperparameter decision is in Appendix A.5.

Table 4.1: Design decisions

Decision	Alternative	Reason
Random Forest	Ridge regression	In site-grouped CV, RF had MAE 0.1159 against 0.1452 for Ridge
Site-holdout as primary evaluation	Random split	A random split measures reproduction of seen sites (Section 2.2)
Keep hyperparameters 50 / 12 / 7	Switch to 100 / 12 / 7	The best leakage-free setting improved site-holdout MAE by only 0.0002, so the original was kept
Disclose defects	Drop affected rows	Dropping rows after seeing results would bias the evaluation

4.4 Detailed design

The service loads the trained model and site metadata, builds the feature row for the requested site and date, predicts the crowd score, and maps it to a risk class (Low, Medium or High). Endpoints include prediction, itinerary check and green-site ranking. Destinations outside the 197-site model (142 of 350 catalogued destinations) receive a “not covered” response and no prediction.

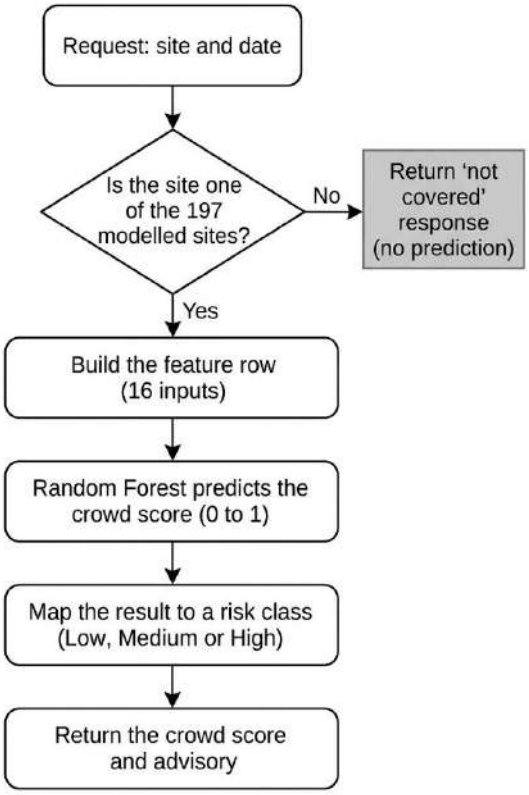


Figure 4.2 Prediction flow

4.5 Standards and professional practice

The code uses fixed environments and version-controlled scripts. Results are labelled by split, and defects and limitations are disclosed.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Python 3.10, scikit-learn 1.7.1, pandas and NumPy for modelling; Flask for the API; React with Vite for the frontend; a unified monorepo for the four components.

5.2 Component implementation

The model is a scikit-learn Random Forest pipeline with one-hot encoding of categorical features. The training script was run with --skip-grid-search, and the retrained model's predictions were identical to the previous pickles. The deployed model was trained on all 197 sites, including the ten held-out sites, so the evaluation results describe performance on unseen sites, not the deployed model's output on the 197 sites it serves.

5.3 Integration with the other components

The risk routes are exposed under /api/risk through risk_mount.py. During integration, a missing mount was found and fixed by creating that file. The SOS tourist frontend was confirmed to call the risk service in the browser's developer tools.

5.4 Verification and software testing

Model verification was carried out by validate_model.py. Its checks and outcomes are in Table 5.1. Known issues at submission are in Table 5.2.

Table 5.1: Verification checks and outcomes

Check	**Outcome**
Reproduce locked site-holdout MAE and R^2	Exact match (0.1232, 0.3331)
Shuffled-target Random Forest	R^2 −0.0146 (grouped CV) and −0.3473 (holdout); no pipeline leak
Unshuffled 5-fold CV inspection	All 197 sites seen in every fold; reported as a

Check	Outcome
	reference only
Read-only check of the model and data folders	No changes

Table 5.2: Known issues at submission

Issue	Effect
/api/risk/green-sites takes about 7 s for 114 eco-sites	Slow response
The /retrain endpoint is a stub that looks for train_model.py at the wrong path	Retraining from the API does not work
142 of 350 catalogued destinations are outside the 197-site model	They receive a “not covered” response
In the standalone repository, the frontend proxy points to port 5002 while the backend runs on 5000	Needs a configuration change

5.5 Deployment and operation

The platform runs as a unified Flask backend on port 5002 with a separate frontend per component.

6 RESULTS AND VALIDATION

6.1 Research results

Table 6.1 gives the primary split. Table 6.2 compares the Random Forest with baselines under every validation design, and Figure 6.1 plots the R^2 values from Table 6.2.

Table 6.1: Primary split (site-holdout)

Item	**Value**
Training sites / rows	187 / 136,697
Held-out sites / rows	10 / 7,310

Table 6.2: Random Forest and baselines, by validation design

Split	**Unseen sites?**	**RF MAE / R^2**	**Best baseline by R^2, MAE / R^2**	**Mean predictor MAE / R^2**
Site-holdout, 10 sites (**primary**)	Yes	**0.1232 / 0.3331**	Category-month 0.2179 / 0.1201	0.2835 / −0.3476
Site-grouped CV, 197 sites, 5×3	Yes	0.1159±0.0133 / 0.5071±0.1090	Ridge 0.1452 / 0.3883	0.1975 / −0.0087
2024→2025, site-grouped CV	Yes	0.1148±0.0109 / 0.5329±0.1004	Ridge 0.1412 / 0.4144	0.1979 / −0.0096
2024→2025, ten held-out sites	Yes	0.1218 / 0.3416	Category-month 0.2179 / 0.1268	0.2837 / −0.3415
2024→2025, same sites	No	0.0709 / 0.8334	Site-month mean 0.0852 / 0.7703	0.1976 / 0.0000
Random 80/20	No	0.0679 / 0.8492	Site-month mean	0.1967 /

Split	Unseen sites?	RF MAE / R^2	Best baseline by R^2, MAE / R^2	Mean predictor MAE / R^2
			0.0863 / 0.7610	0.0000
Unshuffled 5-fold (file order)	No	0.0694±0.0100 / 0.8194±0.0243	n/a	n/a

± is the sample SD across folds. Remaining baselines: category mean 0.1896 / 0.0599 in grouped CV and 0.2273 / 0.0515 on the holdout; Ridge on the holdout 0.2057 / −0.0523. On the two ten-site designs, Ridge has a lower MAE than the category-and-month mean (0.2057 and 0.2056 against 0.2179) but a negative R^2 (−0.0523 and −0.0448). The table lists the baseline with the highest R^2.

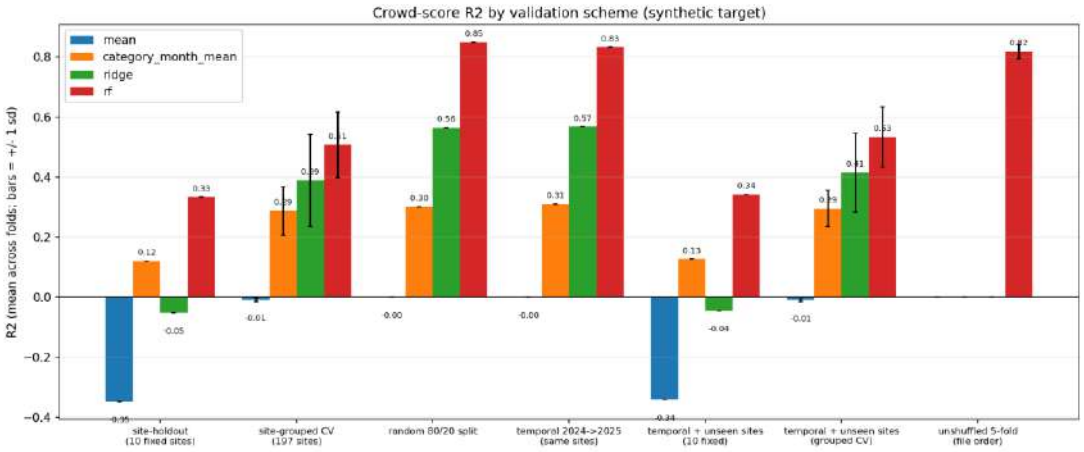


Figure 6.1: R^2 by validation design

On the primary design, the Random Forest cut MAE from 0.2835 (mean predictor) to 0.1232. The lowest-MAE baseline was Ridge at 0.2057, so the Random Forest reduced MAE by about 40% against the strongest baseline, and its R^2 of 0.3331 is well above the best baseline R^2 of 0.1201 (category-and-month mean). Because ten sites are a single draw, site-grouped CV over all 197 sites is the more stable estimate: R^2 0.5071, with fold values from 0.3301 to 0.7060. The ten held-out sites are harder than average, with a mean per-site MAE of 0.1301 against 0.1113 over all 197 sites (out-of-fold). Designs that include the test sites in training reach R^2 0.82 to 0.85, but

a table of per-site, per-month means alone reaches 0.7610 and 0.7703 on the random and temporal splits, so the model's margin there is small.

Classification into Low/Medium/High reached accuracy 0.7731 and balanced accuracy 0.5754 on the holdout, and accuracy 0.7910 and balanced accuracy 0.8000 on the random split. In the random split (n = 28,802), High-class precision was 0.57 and recall 0.87, Medium 0.67 and 0.72, Low 0.95 and 0.81 (Figure 6.2). Most High-class false alarms were Medium days (1,728 of 5,295 days predicted High), and only 1 of 3,510 actual High days was predicted Low.

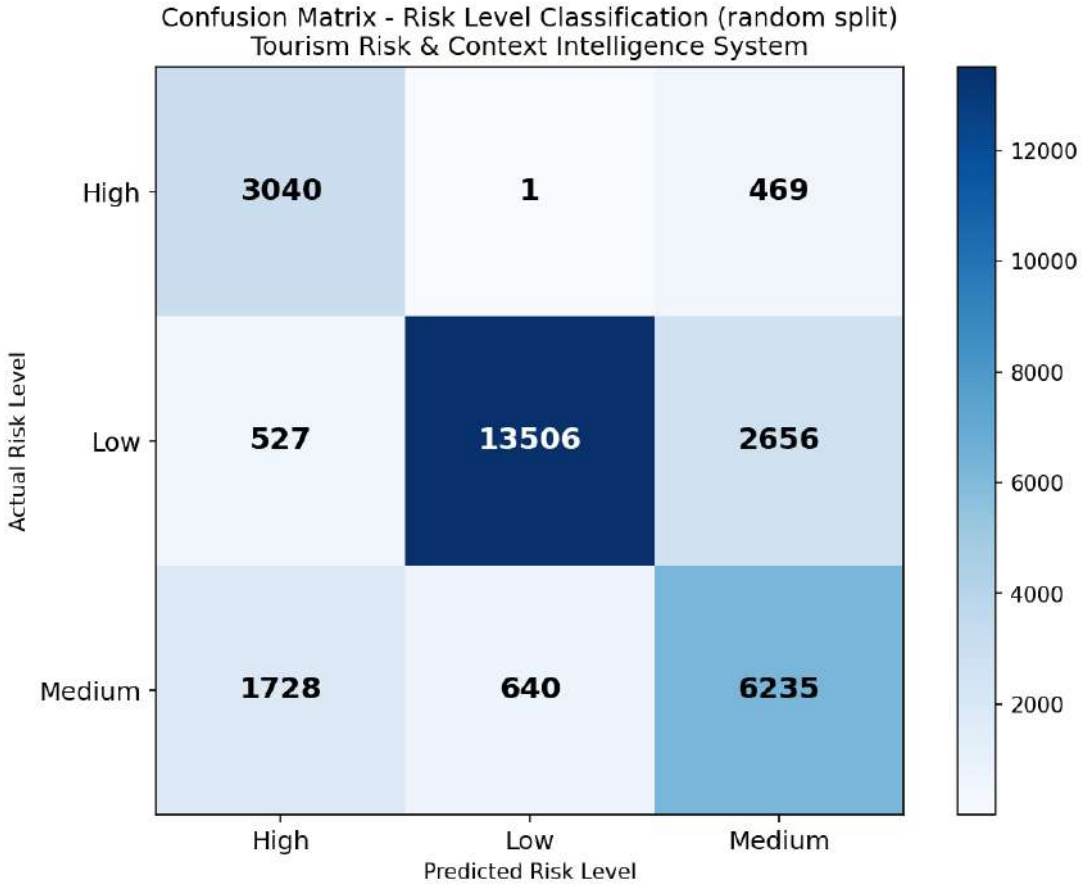


Figure 6.2: Risk-class confusion matrix, random 80/20 split (sites seen in training, so optimistic)

Table 6.3 shows the per-site error for the ten held-out sites, and Table 6.4 the ablation.

Table 6.3: Per-site error, ten held-out sites (out-of-fold, site-grouped CV, repeat 0; 731 days each)

Site	Category	Actual mean	Predicted mean	Bias	MAE
Wilpattu National Park	Nature	0.7050	0.1118	−0.5931	0.5931
Kandy Lake	Nature	0.1118	0.3633	+0.2514	0.2514
Colombo Lotus Tower	Heritage	0.1483	0.2655	+0.1173	0.1173
Minneriya National Park	Nature	0.7040	0.6153	−0.0886	0.1162
Bundala National Park	Nature	0.4953	0.5506	+0.0552	0.0770
Pasikuda Beach	Beach	0.1028	0.1457	+0.0430	0.0430
Kande Viharaya	Religious	0.2258	0.2593	+0.0335	0.0418
Kalpitiya Beach	Beach	0.1258	0.1462	+0.0204	0.0238
Bentota Beach	Beach	0.1362	0.1231	−0.0131	0.0200
Tangalle Beach	Beach	0.1196	0.1277	+0.0081	0.0170

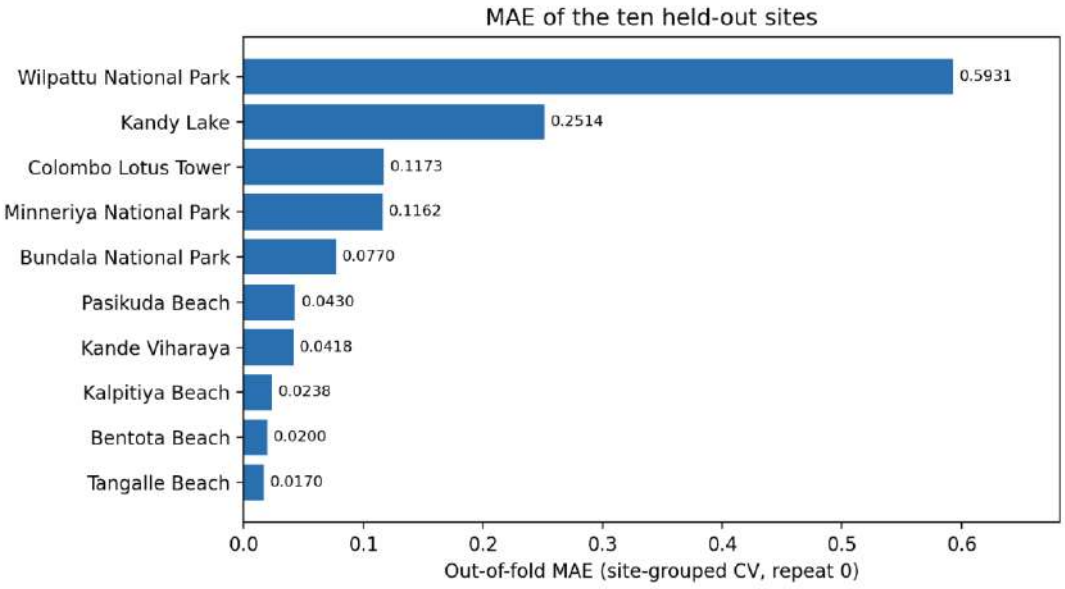


Figure 6.3 per site mae

Source: validation_per_site.csv (out-of-fold predictions, site-grouped CV, repeat 0; 731 days each), rounded to four decimals. These are cross-validated predictions, not outputs of the 187-site holdout model.

Table 6.4: Ablation (RF)

Variant	**Holdout MAE / R^2**	**Grouped-CV MAE / R^2**
Full	0.1232 / 0.3331	0.1113 / 0.5552
Drop occupancy + flights	0.1212 / 0.3429	0.1104 / 0.5571
Drop capacity_per_day	0.2362 / −0.3630	0.1513 / 0.2950
Drop is_weekend + is_public_holiday	0.1228 / 0.3359	0.1116 / 0.5536
Drop season	0.1228 / 0.3385	0.1117 / 0.5531
Drop all of the above	0.2316 / −0.3043	0.1533 / 0.2723
Shuffled target	0.2833 / −0.3473	0.1979 / −0.0146

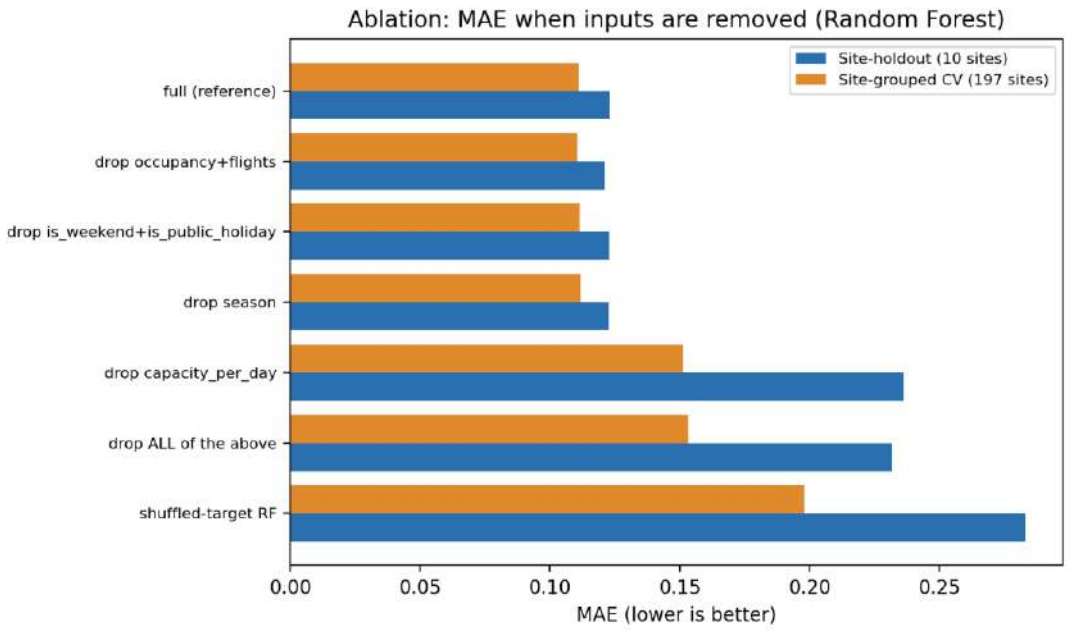


Figure 6.4 ablation mae

6.2 Product and technical results

The risk endpoints respond through the unified backend. The green-site ranking takes about 7 seconds for 114 eco-sites (Table 5.2). Destinations outside the model receive a “not covered” response.

6.3 User or stakeholder validation

At the time of writing, no structured user acceptance testing (UAT) of the crowd component has been completed. The results above therefore evaluate model accuracy against the constructed index. They do not show that users find the advisory useful, and Section 8.4 lists this as the first piece of future work.

6.4 Integration results

The risk routes are served under /api/risk in the unified backend, and the SOS tourist frontend was confirmed to call them (Section 5.3).

6.5 Summary of findings

Table 6.5 sets the proposal targets against the results.

Table 6.5: Proposal targets against results

Target	Result	Met?
MAE ≤ 0.25	0.1232 (site-holdout)	Yes, but Ridge already scores 0.2057 and a category-and-month mean 0.2179, so the target was easy
$R^2 \geq 0.7$	0.8194–0.8492 on seen-site designs; 0.3331–0.5329 on every unseen-site design	Only when sites are seen
Alert precision and recall ≥ 0.80	High class: precision 0.57, recall 0.87 (random split); holdout balanced accuracy 0.5754	No

The model learns site level and category well, learns less about new sites, and gains nothing measurable from occupancy or flight features.

7 DISCUSSION

7.1 Interpretation of findings

What the model learned. Removing capacity per day cut grouped-CV R^2 from 0.5552 to 0.2950 and made the site-holdout R^2 negative (−0.3630). Occupancy and flights, the weekend and holiday flags, and season each changed holdout MAE by 0.0020 or less. The skill therefore comes mainly from a site-level baseline tied to capacity, not from the temporal and contextual features that the platform presents to tourists. This is expected from the construction of the target (Appendix A.1): the generator uses capacity, season, weekend and holiday terms, and never uses hotel occupancy or flight arrivals, so those two inputs could not carry signal about it.

Where it fails. Out-of-fold, the RF beat the category-and-month mean in every category (Table 7.1). It was worse than that baseline at 36 of 197 sites (18.3%). Ten sites had MAE between 0.30 and 0.59 (Wilpattu, Gal Oya, Maduru Oya, Dolukanda, Sigiriya, Lahugala Kitulana, Anawilundawa, Horagolla, Hummanaya and Yala; Appendix A.3). For these, bias equals MAE, so the error is a whole-site offset, not noise. Eight of the ten are Nature sites. Wilpattu contributed about 46% of the summed per-site MAE of the ten held-out sites (0.5931 of 1.3006, Table 6.3), and the other nine average about 0.079. The low primary R^2 is therefore largely a one-site result.

Table 7.1: Out-of-fold error by category

Category	Sites	RF MAE	Category-month mean MAE	Pooled R^2
Beach	26	0.0555	0.1787	0.8851
Heritage	43	0.1190	0.1466	0.5103
Nature	87	0.1375	0.1795	0.4232
Religious	41	0.0832	0.1126	0.5715

Calibration. Across deciles of predicted score (Appendix A.4), mean predictions track observed means within 0.05 in nine of ten bins. The lowest decile under-states

(0.1135 predicted, 0.1890 observed) and the top decile over-states (0.8431 against 0.8090).

Tuning. The best depth of 12 sat at the edge of the search grid, so deeper trees were not tested (Appendix A.5).

7.2 Comparison with literature baselines and existing solutions

The point of comparison is the evaluation design. Random-split R^2 of 0.8492 and site-holdout R^2 of 0.3331 describe the same model, which illustrates the over-optimism that Roberts et al. [2] describe for grouped data. Against simple baselines on the same split, the RF is clearly better on unseen sites (Table 6.2).

7.3 Research contribution

The thesis contributes a defect-audited 197-site dataset, an evaluation protocol that reports every result by split, a measured account of how much a random split overstates performance, and an ablation showing which inputs carry signal.

7.4 Product contribution

A working crowd and risk service, exposed under /api/risk, with an itinerary check, a green-site ranking and a “not covered” response for destinations outside the model.

7.5 Limitations and threats to validity

1. **Constructed target.** Every result measures how well the index is reproduced, not real crowding. This is the main threat to validity. The 2024 and 2025 per-site means correlate at 0.9996 because the generator uses the 2024 annual visitor figure for both years; this is a property of the construction, not evidence of stability.
2. **Capacity dominance.** The target is estimated visitors divided by capacity per day, so capacity is an ingredient of the target’s own formula, and the model’s dependence on it is partly built in. Annual visitor figures are not a model input. Capacity per day is derived from annual visitors for 111 sites.

3. **Uninformative inputs.** Weather depends only on the month, and hotel occupancy and flights are identical across sites, so they cannot carry site-level information.
4. **Single hold-out draw.** Ten sites are one draw; site-grouped CV addresses this.
5. **Few sites drive error.** Wilpattu and a few Nature sites dominate the error.
6. **No user validation** (Section 6.3).
7. **Train/serve skew** in the holiday and festival flags (Section 1.7).

7.6 Implications

For a planner, the service is usable as a relative guide for site type and capacity, not as a precise forecast. Reporting by split lets future work check whether a better target or data source improves unseen-site performance.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

A Random Forest can predict a constructed daily crowd index for unseen sites better than simple baselines (MAE 0.1232 against 0.2057 for Ridge, the lowest-MAE baseline), but only moderately well (R^2 0.3331 on the primary holdout; 0.5071 across 197 sites). The much higher random-split R^2 of 0.8492 reflects sites seen in training. Capacity per day is the main source of skill, and occupancy and flight inputs contribute none. Because the target is constructed, these results do not show real crowding.

8.2 Achievement of objectives

Table 8.1 summarises the outcome for each objective.

Table 8.1: Objectives and outcomes

Objective	**Outcome**	**Evidence**
1 Dataset and defects	Achieved	Section 3.2, Table 3.1
2 Constructed index	Achieved; validity limits stated	Sections 3.2, 7.5, Appendix A.1
3 Train RF and compare baselines	Achieved	Table 6.2
4 Evaluate on unseen sites and time	Achieved	Tables 6.2–6.4
5 REST API under /api/risk	Achieved; known issues in Table 5.2	Sections 5.3–5.4
6 Assess usefulness with users	Not achieved at the time of submission	Section 6.3

8.3 Individual research and product contributions

The research contribution is the leakage-aware evaluation and the measured gap between seen-site and unseen-site performance. The product contribution is the /api/risk service. Details are in Sections 7.3 and 7.4.

8.4 Future directions

1. Run a task-based UAT with a System Usability Scale (SUS) questionnaire [10] and record completion, time and errors.
2. Check the index against observed visitor counts from site operators. A pre-registered comparison script, real_world_check.py, is written but has not been run on real observations. It needs at least 30 observer ratings on a 1 to 5 scale.
3. Replace the shared occupancy and flight series with site-specific data, or remove them.
4. Investigate the Wilpattu and Kandy Lake offsets and the capacity derivation for 111 sites.
5. Search deeper trees, since the best depth sat at the grid edge.
6. Fix the /retrain endpoint, speed up /api/risk/green-sites, and align the training and serving holiday flags.

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GLOSSARY

Ablation – removing an input to see how much the result depends on it.

Balanced accuracy – the mean of per-class recall.

Constructed index – a crowd score computed by project code, not measured.

Shuffled-target test – training on randomly permuted targets, which should score about zero.

Site-grouped cross-validation – folds split by site so that no site appears in both training and test.

Site-holdout – evaluation on sites withheld entirely from training.

APPENDICES

Appendix A Extended results and crowd-index definition

A.1 Crowd-index definition. The target crowd_score_normalized is generated by simulate_crowd_data.py; it is not measured. For each site and day, expected visitors are the site's 2024 annual visitor figure divided by 365, multiplied by a season factor (1.6 for December to March, 1.0 for April to June, 0.8 for July to September, 1.1 for October and November), a weekend factor of 1.4, a public-holiday factor of 1.8 and a noise factor drawn from a normal distribution with mean 1.0 and SD 0.15. The product is truncated to an integer, limited to the site's capacity per day, divided by that capacity and rounded to three decimals, so the score lies between 0 and 1. The same 2024 annual figure is used for both 2024 and 2025. NumPy is seeded with 42. Risk classes are High at 0.75 or above, Medium at 0.45 or above, and Low otherwise. Hotel occupancy and flight arrivals are drawn once per day and shared by all sites, and neither enters the formula. Weather is a fixed monthly lookup. The modelling file master_dataset_fixed.csv is the corrected version of this generator's output (Table 3.1).

base_daily = site["annual_visitors"] / 365

weekend_mult = 1.4 if is_weekend else 1.0

holiday_mult = 1.8 if is_holiday else 1.0

noise = np.random.normal(1.0, 0.15)

estimated_visitors = int(base_daily * season_mult * weekend_mult * holiday_mult * noise)

estimated_visitors = max(0, min(estimated_visitors, site["capacity_per_day"]))

crowd_score = round(estimated_visitors / site["capacity_per_day"], 3)

A.2 Unshuffled 5-fold CV (all 197 sites seen in every fold)

Fold	Test dates	MAE	R^2
0	2024-01-01 to 2024-05-26	0.0766	0.8410
1	2024-05-26 to 2024-10-19	0.0576	0.7783
2	2024-10-19 to 2025-03-14	0.0823	0.8189
3	2025-03-14 to 2025-08-07	0.0629	0.8316
4	2025-08-07 to 2025-12-31	0.0674	0.8275

A.3 Five best and ten worst sites by MAE (out-of-fold)

Best	MAE	Worst	MAE
Kirinda Beach	0.0160	Wilpattu National Park	0.5931
Tangalle Beach	0.0170	Gal Oya National Park	0.4953
Bentota Beach	0.0200	Maduru Oya National Park	0.4443
Kalpitiya Beach	0.0238	Dolukanda Rajamaha Viharaya	0.4337
Casuarina Beach	0.0279	Sigiriya Rock Fortress	0.4226
		Lahugala Kitulana National Park	0.4216
		Anawilundawa Wetland Sanctuary	0.3876
		Horagolla National Park	0.3683
		Hummanaya Blow Hole	0.3090
		Yala National Park	0.3036

A.4 Calibration (deciles of predicted score, out-of-fold)

Bin	Mean predicted	Mean actual
0	0.1135	0.1890
1	0.2209	0.2395
2	0.2891	0.3158
3	0.3332	0.3191
4	0.3821	0.3634
5	0.4329	0.4453
6	0.4828	0.4756
7	0.5787	0.5297
8	0.6862	0.6730
9	0.8431	0.8090

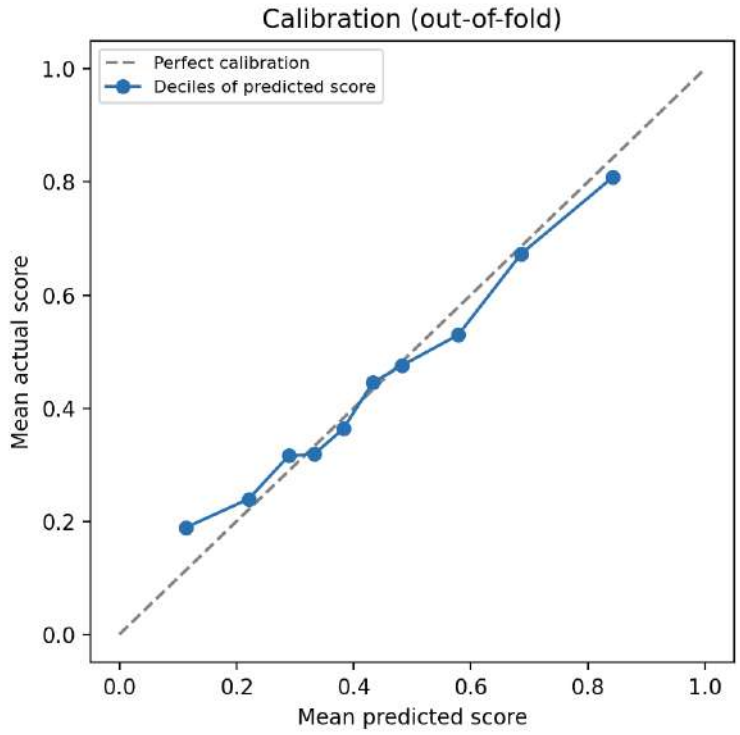


Figure A.1 calibration

A.5 Leakage-free tuning check (top five of 27, site-grouped CV on 187 training sites)

n_estimators	max_depth	min_samples_split	CV MAE	CV R^2
100	12	7	0.1153	0.5239
150	12	7	0.1154	0.5238
100	12	5	0.1154	0.5227
150	12	5	0.1154	0.5227
150	12	3	0.1155	0.5227

The original setting (50 / 12 / 7) ranked 7th of 27. The best setting scored MAE 0.1230 and R^2 0.3351 on the holdout, against 0.1232 and 0.3331 for the original, so the original was kept. Settings are ranked by cross-validated MAE. Depth 10 scores a

marginally higher R^2 (up to 0.5256) at a higher MAE (0.1172 to 0.1173), so the choice of depth 12 follows the MAE ranking.

Appendix B Academic integrity and AI use disclosure

Tool: Claude (Anthropic), via the Claude app.

Purposes: help structuring the thesis against the module template; drafting and rewording text from my own results; explaining evaluation concepts (grouped cross-validation, ablation); commenting on the output of validate_model.py; and writing short read-only scripts that print values from my result files.

My own work: I built the dataset, the crowd index, the model, the validation script and the API, and I ran all code myself. The AI did not run code, produce any number, collect data or build any part of the software. Every number in Chapters 6 and 7 comes from files I produced: the saved outputs of validate_model.py (run on 6 October 2026), the model metadata file, the confusion matrix from the training script, and the dataset.

Verification: on 10 October 2026 I re-checked the numbers in the thesis against those files by running read-only scripts and reading the confusion matrix. I checked each reference before submission.

Sections affected: structure and wording of Chapters 1–8, the appendix tables and the abstract.